

Gokul Elangovan

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EDUCATION

Rutgers University

Bachelor of Science in Computer Science & Data Science

Sept. 2023 - Jan. 2027

GPA: 3.88/4.00

- **Coursework:** Deep Learning, Machine Learning, Systems, Computer Architecture, Data Structures, Algorithms, Data Science, Data Management, Linear Algebra, Multivariable Calculus, Statistics

TECHNICAL SKILLS

Languages: Python, SQL, Java, C, Rust, JavaScript
Libraries: PyTorch (CUDA, MPS), HF Transformers, Scikit-learn, Pandas, NumPy, Matplotlib, Flask, FastAPI
Databases: Google Cloud Storage, BigQuery, PostgreSQL, SQLite, MongoDB
Tools: Git, GitHub, Docker, Kubernetes, Vertex AI (Gemini, embeddings), GCP, Linux/Ubuntu

EXPERIENCE

Wayfair

Software Engineer Intern

June 2026 - Aug. 2026

Boston, MA

- Minimized **Gemini-2.5-pro** chat context payloads in **Wayfair Intelligence System**, cutting chat input tokens **55% (970k to 433k)** on average, validated by a harness instrumenting per-call latency, tokens, and cost
- Prototyped a **RAG** pipeline with **Gemini text-embedding-004** using **cosine retrieval** and full-context fallback, reducing average chat call cost from **\$0.54 to \$0.02**, and input tokens from **433k to 14k**
- Shipped **LRU & TTL** report cache, saving **~\$0.70** and **~70 seconds** per repeat request deployed via K8s
- Built end-to-end saved-reports feature: **GCS** bundle storage (report HTML, context, manifest), **BigQuery** index for listing/search, and **Flask REST API** with **IAP auth** enforcing per-user isolation across WINS

Rutgers University

Undergraduate Researcher

June 2025 - Present

New Brunswick, NJ

- Built a CLIP test-time adaptation pipeline (TPT, TDA) benchmarked under synthetic domain shifts, finding that a per-image strategy router has **~6 points of headroom (69.8% oracle vs. 64.0% best fixed strategy)**
- Trained gating models (gradient boosting and MLP), **beating always-TPT selection by ~4-5 points**; evaluated RL-based routing, diagnosing majority-class collapse, and a **12x compute gap versus TPT/TDA**

Transatlantic Reinsurance

Data Science Intern

June 2025 - Aug. 2025

New York, NY

- Developed a **high-throughput ETL pipeline** processing **1,200+ wildfire records spanning 20 years**, forming the basis for **20+ automated Excel dashboards** delivered by year for actuarial risk modeling
- Identified **6 wildfire trends** using regression, clustering, and visualization in scikit-learn and Matplotlib

CodePath

Tech Fellow - AI 110, TIP 101/102

July 2024 - Present

Remote

- Instructed **600+ students** across **3 cohorts** in AI110, covering AI engineering with Claude Code and Python
- Instructed **1,000+ students** across **5 cohorts** in TIP101, covering data structures and algorithms in Python

PROJECTS

Chain of Thought (CoT) Knowledge Distillation (KD)

- Built **CoT distillation** pipeline in **PyTorch**, with **sequence** and **logit-level KD** with cross-tokenizer alignment via character-offset tracking from **70B+ param teachers (Llama 3.3, GPT-OSS)** and **Llama 3.2 3B student**
- Identified correctness vs. style tradeoff; **logit KD outperformed baseline by 29%**, and **sequence KD** achieved the best teacher replication with a **0.9693 F1 BERTScore** across SciBench and TheoremQA

Mixture-of-Experts LLM

- Designed a custom **MoE decoder-only transformer** in **PyTorch**; implemented **Top-k expert routing** with **loss-free load balancing** to optimize compute unit utilization and reduce inference latency across experts
- Implemented **GQA** and **RoPE** to **reduce KV-cache memory bandwidth**; applied **weight tying** across token embeddings and output projection to minimize memory footprint and parameter count

LEADERSHIP/EXTRACURRICULAR

Rutgers Undergraduate Student Alliance of Computer Scientists (USACS)

Tech Director, Tech Chair

Jan. 2025 - May 2026

- Directed UHACCS hackathon with **150+ participants**, securing partnerships with **Microsoft** and **Figma**